Applying Performance Optimization Techniques in Zoho Creator

Aim

To apply performance optimization techniques such as indexing, query tuning, and subforms in a Zoho Creator application to improve data retrieval speed, reduce unnecessary processing, and organize related data efficiently.

Procedure

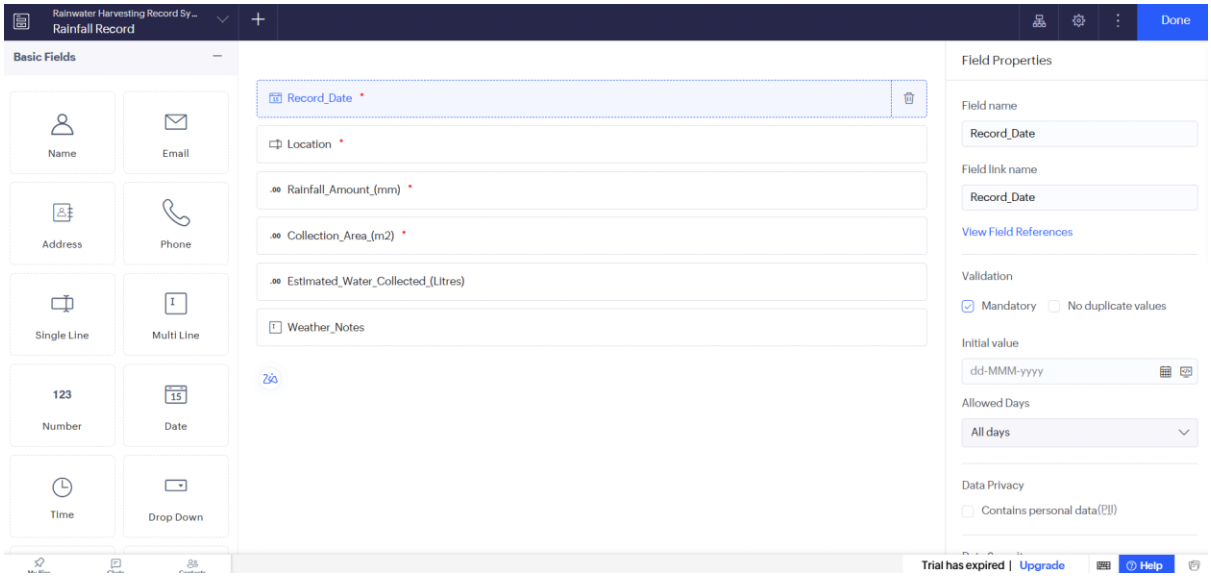
1. Indexing
    1. Open the Rainwater Harvesting Record System in Zoho Creator.
    2. Identify fields frequently used for searching, filtering, and sorting, such as Record Date and Location in the Rainfall Record form.
    3. Check the available field and indexing options in the application settings.
    4. If indexing is supported for the selected field and plan, configure it appropriately.
    5. Test data retrieval before and after optimization, if possible.
2. Query Tuning
    1. Open the reports associated with the application.
    2. Review the filters used in the Rainfall Analysis Report, Storage Tank Status Report, and Water Usage Report.

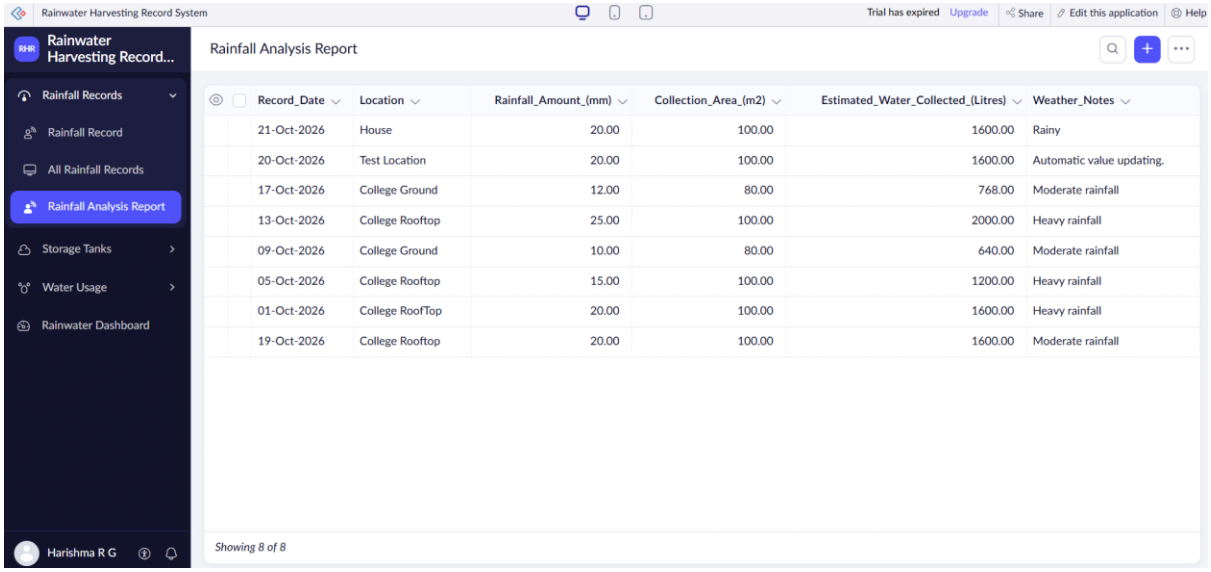
3. Apply relevant filters to retrieve only the required records.
4. Avoid unnecessary data retrieval and repeated processing in Deluge scripts.
5. Test the reports with sample records and observe their response time.

3. Subforms
    1. Open the application builder and select a suitable form, such as Storage Tank or Water Usage.
    2. Edit the form and check whether a subform can be added.
    3. Add a subform to maintain multiple related entries within a single parent record, if required.
    4. Configure suitable fields for the subform, such as usage date, water quantity, and usage purpose.
    5. Save the form and test adding multiple entries.
    6. Verify that the parent record and its related entries are displayed correctly.
4. Testing and Verification
    1. Enter sample rainfall, storage tank, and water usage records.
    2. Test the reports using different filters.
    3. Verify the organization of related data using the subform, if configured.

4. Compare the results and note any improvements in retrieval speed, usability, and data management.

OUTPUT





Record_Date	Location	Rainfall_Amount_(mm)	Collection_Area_(m2)	Estimated_Water_Collected_(Litres)	Weather_Notes
21-Oct-2026	House	20.00	100.00	1600.00	Rainy
20-Oct-2026	Test Location	20.00	100.00	1600.00	Automatic value updating.
17-Oct-2026	College Ground	12.00	80.00	768.00	Moderate rainfall
13-Oct-2026	College Rooftop	25.00	100.00	2000.00	Heavy rainfall
09-Oct-2026	College Ground	10.00	80.00	640.00	Moderate rainfall
05-Oct-2026	College Rooftop	15.00	100.00	1200.00	Heavy rainfall
01-Oct-2026	College RoofTop	20.00	100.00	1600.00	Heavy rainfall
19-Oct-2026	College Rooftop	20.00	100.00	1600.00	Moderate rainfall

Result

Performance optimization techniques in Zoho Creator were studied using the Rainwater Harvesting Record System. Indexing was considered for frequently accessed fields to support efficient data retrieval where supported. Query tuning through relevant filters helped focus reports on the required records and reduce unnecessary processing. Subforms provided a way to organize multiple related entries under a single parent record.

Thus, the experiment demonstrated how indexing, query tuning, and subforms can contribute to improved application performance, better data organization, and easier report management.